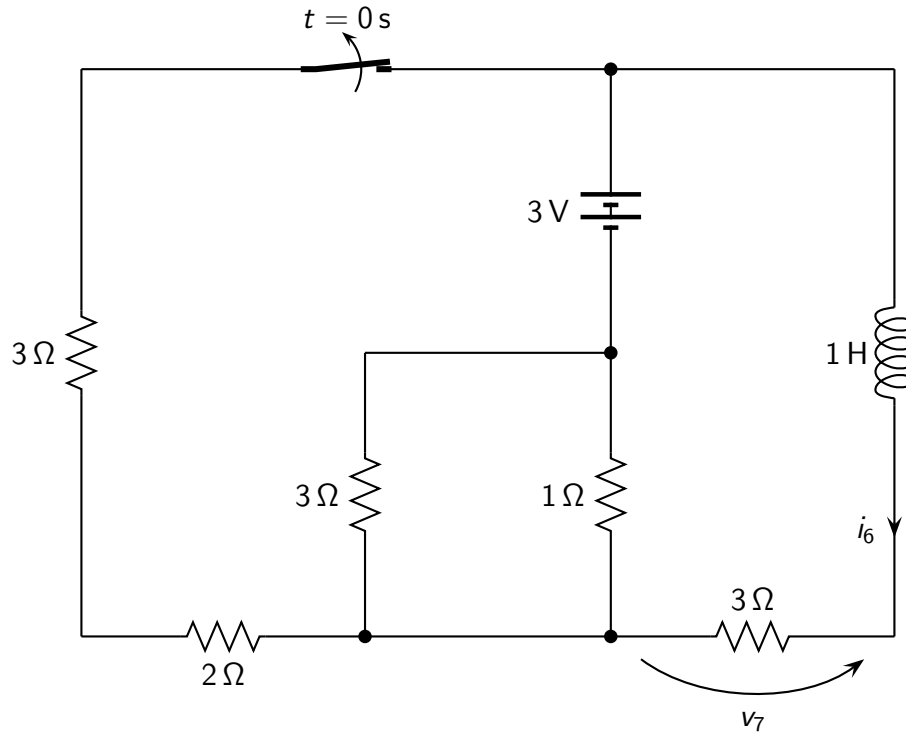


Problem: find the expression of $i_6(t)$, $v_7(t)$ for $t > 0$.



Solution

- Time constant: $\tau = \frac{4}{15} \text{ s}$;
- Initial conditions before switching: $i_6(0^-) = \frac{5}{7} \text{ A}$;
- Initial conditions after switching: $i_6(0^+) = \frac{5}{7} \text{ A}$; $v_7(0^+) = \frac{15}{7} \text{ V}$;
- Steady-state solution: $i_6(\infty) = \frac{4}{5} \text{ A}$; $v_7(\infty) = \frac{12}{5} \text{ V}$;
- Solution for $t > 0$:

$$i_6(t) = \left(\frac{-3}{35} e^{-t/\tau} + \frac{4}{5} \right) \text{ A}$$

$$v_7(t) = \left(\frac{-9}{35} e^{-t/\tau} + \frac{12}{5} \right) \text{ V}$$